

Tuesday, March 24

Welcome back!!
😊

Plan

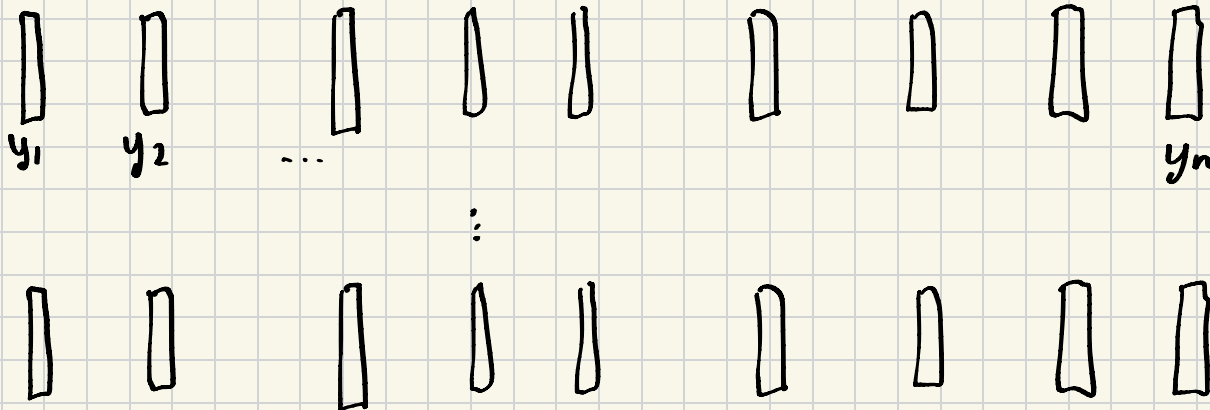
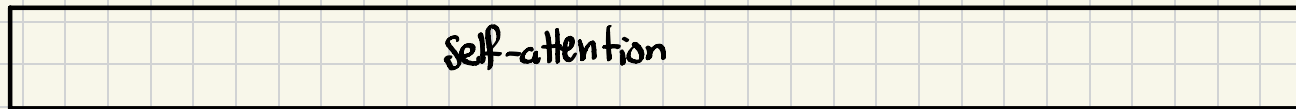
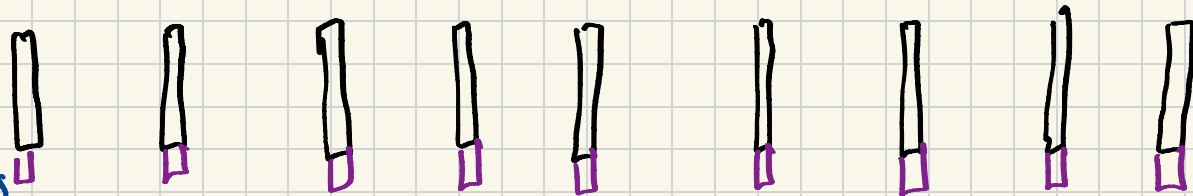
- Positional encoding
- Decision trees

Input: The weather today is warm and sunny.

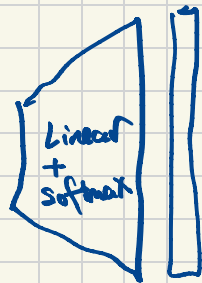
Tokens: "the" "weather" "today" "is" "warm" "and" "sun" "ny" "."

Vectors:

using
autoencoders



dist. over
tokens

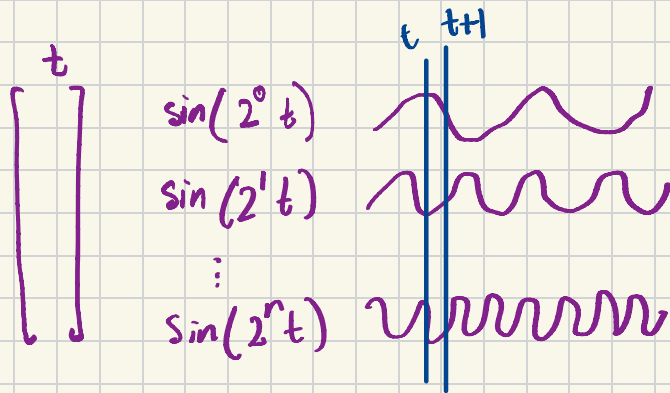
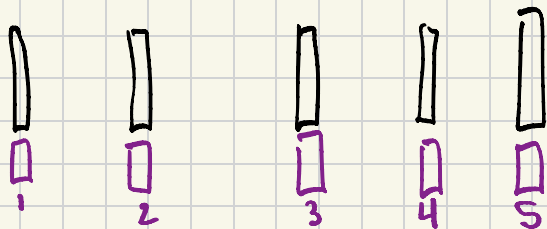


Positional Encoding

"Jack gave water to Jill"

vs.

"Jill gave water to Jack"



Demo on CNN and transformer ☺

Thursday, March 26

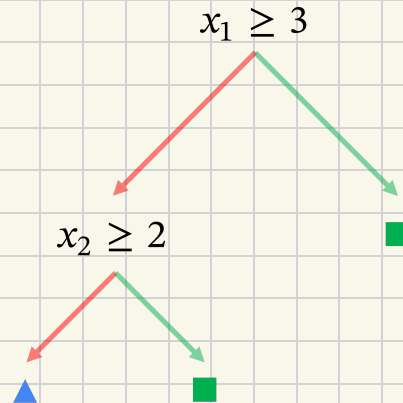
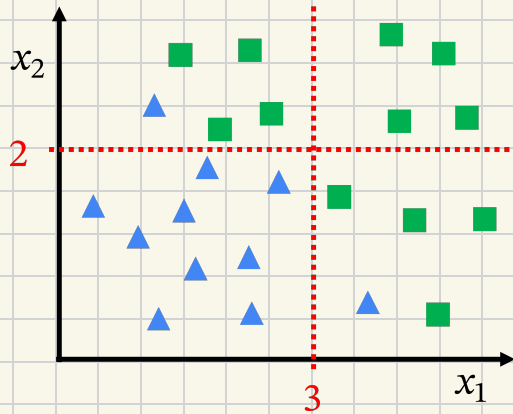
- Shapley value talk by James Gouen (usc) talk on Thursday
(no CH that day)

Today

- Decision trees and boosting!

Decision Trees

- ↳ Remarkably accurate and interpretable
- ↳ works for classification and regression
- ↳ Idea: Recursively partition data into homogeneous regions



Greedy Approach for Classification

For each leaf, choose the feature and split that best separates classes

Let l be candidate leaf and $p_c^{(l)}$ the proportion of points of class c in leaf l .

WLOG, $c=0$ is the majority class

Goal: Find l to minimize

$$\mathbb{E}_l(\mathcal{L}(l)) = \sum_c \mathcal{L}(l) p_c^{(l)}$$

↙ proportion of points in leaf l

↑ How should we define loss?

Error Rate of Majority Prediction:

$$\mathcal{L}_{\text{error}}(\ell) = 1 - p_0^{(\ell)}$$

Gini Impurity: Probability of misclassifying point if we randomly predict based on proportion of class

$$\mathcal{L}_{\text{Gini}}(\ell) = \sum_c p_c^{(\ell)} (1 - p_c^{(\ell)}) = \sum_c p_c^{(\ell)} - \sum_c (p_c^{(\ell)})^2 = 1 - \sum_c (p_c^{(\ell)})^2$$

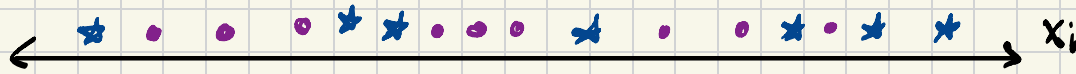
Entropy: like cross-entropy, measure "spread" of distribution

$$\mathcal{L}_{\text{entropy}}(\ell) = - \sum_c p_c^{(\ell)} \log(p_c^{(\ell)})$$

$$\operatorname{argmin}_{\ell} \mathcal{L}_{\text{entropy}}(\ell) = \operatorname{argmax}_{\ell} \prod_c p_c^{(\ell)}$$

Previously we updated parameters to minimize loss;

For decision trees, we search over splits in data to minimize loss.



Time complexity to find best split in feature x_i ?

Q: How do we adapt for regression?

Gradient Boosting

On their own, trees are not very expressive. But we can "boost" them into very strong ensembles.

Let $f_t: \mathbb{R}^d \rightarrow \{-1, 1\}$ is the weak learner from iteration t

$$F_t(x) = \alpha_1 f_1(x) + \dots + \alpha_{t-1} f_{t-1}(x) + \alpha_t f_t(x) = F_{t-1}(x) + \alpha_t f_t(x).$$

Goal: Given F_{t-1} , find f_t to minimize:

$$\argmin_{f_t} \sum_{i=1}^n \mathcal{L}(y^{(i)}, F_{t-1}(x^{(i)}) + \alpha f_t(x^{(i)}))$$

Goal: Given F_{t-1} , find f_t to minimize:

$$\operatorname{argmin}_{f_t} \sum_{i=1}^n \mathcal{L}(y^{(i)}, F_{t-1}(x^{(i)}) + \alpha f_t(x^{(i)}))$$

Idea: Gradient descent on predictions rather than parameters.

$$F_t(x^{(i)}) \leftarrow F_{t-1}(x^{(i)}) - \alpha \frac{\partial \mathcal{L}(y^{(i)}, F_{t-1}(x^{(i)}))}{\partial F_{t-1}(x^{(i)})}$$

Train f_t so that

$$f_t(x^{(i)}) \approx - \frac{\partial \mathcal{L}(y^{(i)}, F_{t-1}(x^{(i)}))}{\partial F_{t-1}(x^{(i)})} \quad \text{for all } i$$

When loss is MSE: $\mathcal{L}(y, F_{t-1}(x)) = \frac{1}{2} (F_{t-1}(x) - y)^2$, then

$$- \frac{\partial \mathcal{L}(y^{(i)}, F_{t-1}(x^{(i)}))}{\partial F_{t-1}(x^{(i)})} = y - F_{t-1}(x) = \text{residual error of } F_{t-1}$$

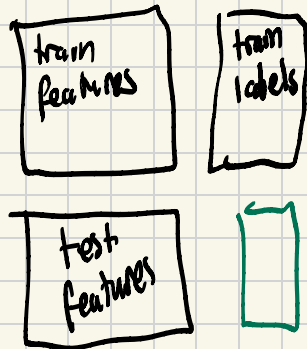
Choose learning rate α , when $\mathcal{L}$ convex,

$$\alpha_t = \operatorname{argmin}_{\alpha} \sum_{i=1}^n \mathcal{L}(y^{(i)}, F_{t-1}(x^{(i)}) + \alpha f_t(x^{(i)}))$$

by setting derivative to 0 and solving for α

XGBoost was SOTA on tabular data from 2014-2025

TabPFN ^{and friends} (Transformer for tabular data) is now SOTA



↳ in context learning